The negative-parameter ACMS reciprocal-tail formulas

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Abstract

We prove the reciprocal-tail floor formula for every pair of integers $s > t \geq 1$, except $(s,t) = (2,1)$, and every positive starting index and step. We also prove the reciprocal-square formula for every integer $s \geq 3$ and every positive starting index and step. The squared formula as printed needs this restriction: $s = 1$ gives zero denominators and $s = 2$ gives counterexamples. The proofs give strict bounds without any large-parameter qualification.

Task 1. Complete resolution

Write $u_j = f_j(s, -t)$, so

$$u_0 = 0, \qquad u_1 = 1, \qquad u_j = su_{j-1} - tu_{j-2} \quad (j \geq 2).$$

The question is Conjecture 2 in Park [1, p. 8], originally Conjecture 4.6 of Amdeberhan–Chen–Moll–Sagan [2, p. 556]. Park's Theorem 18 establishes large-parameter cases; the final ACMS paper's note on p. 560 records that partial result.

Theorem 1. *Let s, t be integers with $s > t \geq 1$ and $(s,t) \neq (2,1)$. For all positive integers n, r, put*

$$F_k = u_{rk}, \qquad S_n = \sum_{k=n}^{\infty} \frac{1}{F_k}, \qquad \Delta_n = F_n - F_{n-1}.$$

Then the series converges and

$$\Delta_n - 1 < S_n^{-1} < \Delta_n.$$

Consequently $\lfloor S_n^{-1} \rfloor = F_n - F_{n-1} - 1$.

Theorem 2. *Let $s \geq 3$ be an integer, take $t = 1$, and let $n, r \geq 1$. With $F_k = f_{rk}(s, -1)$, set*

$$Q_n = \sum_{k=n}^{\infty} \frac{1}{F_k^2}, \qquad \Gamma_n = F_n^2 - F_{n-1}^2.$$

Then the series converges and

$$\Gamma_n - \frac{2}{s^2 - 4} < Q_n^{-1} < \Gamma_n.$$

In particular $\lfloor Q_n^{-1} \rfloor = F_n^2 - F_{n-1}^2 - 1$.

Root estimates and the determinant identity

Under the hypotheses of Theorem 1, $s \geq 3$ and $t \leq s-1$. Define

$$D = s^2 - 4t, \quad \delta = \sqrt{D}, \quad \alpha = \frac{s+\delta}{2}, \quad \beta = \frac{s-\delta}{2}.$$

Since $D \geq (s-2)^2$, we have

$$\delta \geq 1, \qquad \alpha \geq s-1 \geq 2, \qquad 0 < \beta \leq 1, \qquad \alpha\beta = t.$$

Binet's formula, verified by the recurrence and initial conditions, is

$$u_j = \frac{\alpha^j - \beta^j}{\delta}.$$

Thus $0 < u_j < \alpha^j/\delta$ for $j \geq 1$. If $j, h \geq 1$, then

$$\frac{u_j}{u_{j+h}} = \alpha^{-h} \frac{1-(\beta/\alpha)^j}{1-(\beta/\alpha)^{j+h}} < \alpha^{-h}. \tag{1}$$

The inequality remains true for $j = 0$, since $u_0 = 0$. In particular $F_{k+1} > \alpha^r F_k$ for $k \geq 1$, which proves convergence of both series used above.

We shall use the following form of the generalized Cassini identity, also stated as ACMS Lemma 4.3 and Park Lemma 5:

$$F_n F_{n+m-1} - F_{n-1} F_{n+m} = t^{r(n-1)} u_r u_{rm} \quad (m, n \geq 1). \tag{2}$$

For completeness, let $A = \alpha^r$, $B = \beta^r$. Substituting $F_k = (A^k - B^k)/\delta$ into the left side gives

$$\frac{(AB)^{n-1}(A-B)(A^m - B^m)}{\delta^2},$$

which is exactly the right side. In particular it is strictly positive.

The reciprocal formula

Fix $n, r \geq 1$. Define, for $m \geq 1$,

$$C_m = \frac{F_n}{F_{n+m}} - \frac{F_{n-1}}{F_{n+m-1}}.$$

Identity (2) gives $C_m > 0$, and absolute convergence allows us to shift indices to obtain the exact equality

$$\Delta_n S_n = 1 + \sum_{m=1}^{\infty} C_m. \tag{3}$$

Moreover, (2), (1), and the bound on u_r give

$$\begin{aligned} C_m F_{n+m-1} &= t^{r(n-1)} u_r \frac{u_{rm}}{u_{r(n+m)}} \\ &< t^{r(n-1)} \frac{\alpha^r}{\delta} \alpha^{-rn} = \frac{\beta^{r(n-1)}}{\delta} \leq 1. \end{aligned}$$

Summing the strict positive inequalities $0 < C_m < 1/F_{n+m-1}$ in (3) yields

$$1 < \Delta_n S_n < 1 + S_n.$$

Division by $S_n > 0$ proves Theorem 1. This includes $n = 1$, where $F_{n-1} = 0$, and the boundary $t = s-1$, where $\beta = 1$: the preceding strict inequalities still apply.

The reciprocal-square formula

Now take $t = 1$, $s \geq 3$. Thus $D = s^2 - 4 \geq 5$. For $m \geq 1$ put

$$E_m = \left(\frac{F_n}{F_{n+m}}\right)^2 - \left(\frac{F_{n-1}}{F_{n+m-1}}\right)^2.$$

The two ratios are nonnegative and their difference is positive by (2); hence $E_m > 0$. As before,

$$\Gamma_n Q_n = 1 + \sum_{m=1}^{\infty} E_m. \tag{4}$$

Factoring the difference of squares and using (2) with $t = 1$ gives

$$E_m F_{n+m-1}^2 = u_r u_{rm} \frac{F_{n+m-1}}{F_{n+m}} \left(\frac{F_n}{F_{n+m}} + \frac{F_{n-1}}{F_{n+m-1}}\right).$$

Estimate (1) applies to every ratio here, including the zero numerator when $n = 1$. Therefore

$$0 < E_m F_{n+m-1}^2 < \frac{\alpha^r}{\delta} \frac{\alpha^{rm}}{\delta} \alpha^{-r}\left(2\alpha^{-rm}\right) = \frac{2}{D}.$$

Summing in (4) gives

$$1 < \Gamma_n Q_n < 1 + \frac{2}{D} Q_n.$$

This is the claimed strict interval. Since $2/D \leq 2/5 < 1$ and Γ_n is an integer, Theorem 2 follows.

The omitted small parameters in the printed squared formula

Both cited statements print the squared formula for positive s without excluding $s = 1, 2$. If $s = 1$, the recurrence has period six, beginning $0, 1, 1, 0, -1, -1$. Every subsequence u_{rk} contains zeros at arbitrarily large indices, so the proposed series is undefined.

If $s = 2$, then $u_j = j$. At $n = 3, r = 1$, strict integral comparison gives

$$\frac{1}{3} = \int_3^{\infty} x^{-2}\, dx < \sum_{k=3}^{\infty} \frac{1}{k^2} < \int_2^{\infty} x^{-2}\, dx = \frac{1}{2}.$$

The floor of the inverse sum is consequently 2, whereas the printed right side is $3^2 - 2^2 - 1 = 4$. Thus the unqualified squared assertion is false. Theorem 2 proves it throughout the intended nondegenerate range; the small-parameter defect is not the basis for claiming a resolution of the substantive problem.

Exact checks and follow-up tasks

The companion verifier uses integer recurrence values and rational intervals for infinite tails. The bound $F_{k+1} > 2^r F_k$ implies that the tail after K is less than $1/(F_K^p(2^{rp} - 1))$ for $p = 1, 2$. These finite checks corroborate the proof; the argument above establishes all parameters and indices.

Task 2. Higher powers. For $p \geq 3$, determine correction terms to $F_n^p - F_{n-1}^p$ that give the exact floor of $(\sum_{k=n}^{\infty} F_k^{-p})^{-1}$ for $t = 1$ and arbitrary step r. The known cubic result of Yuan–He–Zhou [3] treats step one.

Task 3. A wider parameter range. For integer $s, t > 0$ with $s^2 > 4t$ and $\alpha > 1$, classify the pairs for which the formula of Theorem 1 holds for every $n, r \geq 1$, allowing the smaller root β to exceed one.

Task 4. Shifted progressions. Determine exact floor formulas for the inverses of

$$\sum_{k=n}^{\infty} \frac{1}{u_{rk+a}} \quad \text{and} \quad \sum_{k=n}^{\infty} \frac{1}{u_{rk+a}^2},$$

for $a \geq 0$, in the corresponding parameter ranges of the two theorems.

Task 5. Optimal error bounds. For fixed $s \geq 3, r \geq 1$, determine $\sup_{n \geq 1}(\Gamma_n - Q_n^{-1})$ and its dependence on s, r, including whether the supremum is attained or is a limiting value.

References

[1] S. Park, *Arithmetic properties of generalized Fibonacci sequences*, arXiv:1407.8086v1 (2014), Conjecture 2 and Theorem 18.

[2] T. Amdeberhan, X. Chen, V. H. Moll, B. E. Sagan, *Generalized Fibonacci polynomials and Fibonomial coefficients*, Ann. Comb. **18** (2014), 541–562, doi:10.1007/s00026-014-0242-9.

[3] P. Yuan, Z. He, J. Zhou, *On the sum of reciprocal generalized Fibonacci numbers*, arXiv:1503.00803v1 (2015), Theorem 1.1.